QUESTION 1

Start

Input surname

input password

check if surname and password are correct

if credentials are correct then

identify user's role

if role is Administrator then

allow access to all features

else if role is Safety Officer then

allow dashboard

allow worker safety

allow safety incidents

allow risk assessment

allow reports

else if role is Mining Engineer or Manager then

allow dashboard

allow worker safety

allow safety incidents

allow equipment data

allow maintenance data

allow risk assessment

allow reports

else if role is Maintenance Engineer then

allow dashboard

allow equipment data

allow maintenance data

allow reports

end if

show approved modules

else show "Invalid username or password"

end if

stop

QUESTION 2

input worker id

input department

input job role

input shift

input ppe compliance

input safety training status

input fatigue level

input safety observations

input near misses

input previous incidents

input risk level

check worker safety information

if ppe is not compliant or training is incomplete or fatigue is high or risk level is high/critical or previous incidents exist then

flag worker record as high risk

display safety warning

end if

save worker information

stop

QUESTION 3

Start

Input incident id

Input date

Input time

Input shift

Input location

Input department

input incident type

Input severity

input injury status

Input lti status

input root cause

input corrective action

Input incident status

Save incident information

allow user to search incidents

allow user to view incidents

Allow user to filter incidents

Count incidents

Analyse incidents

Stop

QUESTION 4

start

input equipment id

input equipment type

input manufacturer

input operating hours

input temperature

input vibration

input fuel consumption

input brake status

input tyre status

input engine status

input maintenance status

input downtime

input availability

Save equipment information

STOP

QUESTION 5

start

select equipment

input temperature

input vibration

if temperature > 100 then

temperature status = critical

else if temperature >= 80 and temperature <= 100 then

temperature status = warning else temperature status = normal

end if

if vibration > 8 then

vibration status = critical

else if vibration >= 5 and vibration <= 8 then

vibration status = warning

else vibration status = normal

end if

display temperature status

display vibration status

stop

QUESTION 6

start

select equipment

input operating time

input downtime

calculate availability

availability = (operating time / (downtime + operating time)) × 100

check maintenance status

check if maintenance is due or overdue

check excessive downtime

check repeated alerts

check low availability

display maintenance status

stop

QUESTION 7

start

input likelihood from 1 to 5

input consequences from 1 to 5

risk score = likelihood × consequences

if risk score is 1 to 4 then

risk = low

else if risk score is 5 to 9 then

risk = medium

else if risk score is 10 to 16

then risk = high

else if risk score is 17 to 25 then

risk = critical

end if

if risk is high or critical then

display warning

end if

display risk score

display risk classification

stop

QUESTION 8

start

check worker safety information

check equipment conditions

check maintenance information

check safety incidents

if an unsafe condition is found then

create alert

if condition is critical then

mark alert as critical

end if

display alert

end if

continue monitoring

stop

QUESTION 9

Start

load worker data

load safety incident data

Load equipment data

Load maintenance data

calculate kpis

Display total workers

display safety incidents

Display near misses

display high-risk observations

display ppe compliance

Display equipment availability

display maintenance information

Display equipment alerts

Display critical alerts

When new data is available:

update dashboard

Stop

QUESTION 10

start

load mining data

display analysis options

input user's selection

filter data according to selection

analyse selected data

calculate: totals averages percentages

display results

allow user to change filters

repeat analysis until user exits

stop